



The Paradox and Fallacy of Global Carbon Credits: A Theoretical Framework for Strengthening Climate Change Mitigation Strategies

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Abstract

Carbon credit systems have emerged as a key policy tool in global efforts to mitigate climate change. Under these market-based schemes, entities that reduce greenhouse gas emissions below a specified cap are issued tradable credits that can be sold to entities exceeding their emissions allowances. In theory, this creates financial incentives for emissions reductions and allows reductions to occur where they are most cost-effective. However, this Policy Analysis argues that the increasing reliance on global carbon markets as a climate solution is deeply problematic and paradoxical. Drawing on interdisciplinary perspectives from environmental economics, law, ethics, and political ecology, I explore the problem through a theoretical framework elucidating the inherent limitations and fallacies underpinning carbon credit approaches. I contend that carbon markets (1) fail to produce real, verifiable emissions reductions; (2) exacerbate global inequities and undermine climate justice; (3) distort climate policy by emphasising cost-effectiveness over ecological necessity; and (4) delay the structural economic changes needed to achieve deep decarbonisation. My analysis is supported by a narrative review of the literature, case studies, and empirical evidence. This Policy Analysis concludes by proposing alternative climate mitigation approaches that could help transcend the paradoxes of global carbon credits, including accelerating fossil fuel phaseouts, scaling up public investment in low-carbon infrastructure, advancing climate justice principles, and rethinking carbon accounting frameworks.

Article Highlights

- Carbon credit systems have emerged as a key policy tool for mitigating climate change, but their effectiveness and equity are increasingly being questioned.
- This Policy Analysis explores the problem through a theoretical framework that elucidates four major paradoxes and limitations of carbon markets.
- The analysis draws on interdisciplinary perspectives and is supported by a narrative review of the literature, case studies, and empirical evidence.
- The Policy Analysis proposes alternative climate mitigation approaches that could help transcend the paradoxes of global carbon credits.
- Developing effective and equitable climate change mitigation policies in the coming decades will require moving beyond the limitations of carbon markets.

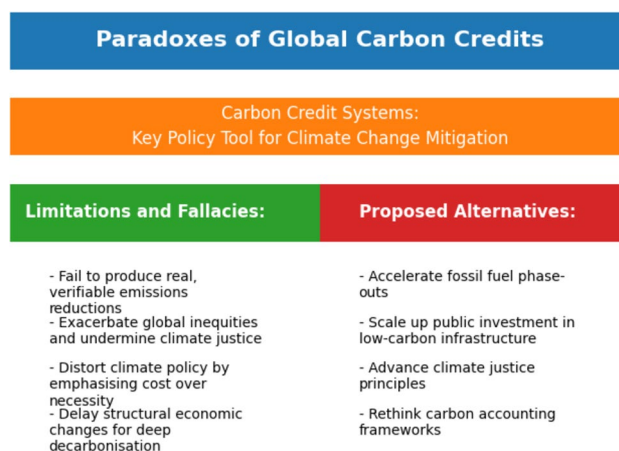
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Graphical Abstract



Keywords Carbon credits · Climate change mitigation · Environmental policy · Climate justice · Low-carbon transition · Green investment · Fossil fuel phaseout · Carbon accounting

1 Introduction

Figure 1 provides a visual representation of the key concepts and their relationships discussed in this Policy Analysis. The diagram highlights the central role of carbon markets (blue circle) in global climate governance (green circle) since their integration into international climate agreements and national emissions reduction strategies in the 1990 s (Kolk et al. 2008; Lovell 2010). However, as the critical scholarship (red circle) increasingly challenges the effectiveness and equity of carbon markets (Lohmann 2009; Böhm et al. 2012; Bryant 2016), this Policy Analysis (yellow circle) contributes by exploring the problem through a theoretical framework that elucidates the paradoxes and fallacies underpinning carbon credit approaches. The arrows connecting the circles illustrate the interconnectedness of these concepts and the flow of ideas between them, with carbon markets influencing climate governance,

while critical scholarship and the Policy Analysis critique and propose alternatives to carbon markets.

Market-based instruments have become central to transnational environmental governance in recent decades (Bernstein et al. 2010). Carbon credit systems are a prominent example, allowing entities that reduce greenhouse gas (GHG) emissions below a specified cap to sell surplus reductions as ‘carbon credits’ to entities exceeding their cap. Since the 1990s, carbon credit approaches have been integrated into international climate agreements and national emissions reduction strategies (Kolk et al. 2008; Lovell 2010; Cheong 2024). The Kyoto Protocol established the first global carbon markets in 1997, which were further expanded in the Paris Agreement (Ciplet 2015). Today, carbon is traded as a commodity in complex compliance and voluntary markets worldwide (Knox-Hayes 2016).

Proponents argue that carbon credits harness market efficiencies to reduce emissions cost-effectively (Stern 2007; Gómez-Baggethun and Muradian 2015). Putting a price on carbon aims to incentivise low-carbon practices and technologies (Bumpus and Liverman 2008). The fungibility of credits allows emissions reductions where they are cheapest, such as through industrial efficiency, preventing deforestation, or renewable energy (Boyd et al. 2011). Carbon markets have thus been framed as a ‘win–win’ solution for the climate and economy (Newell and Paterson 2010).

However, critical scholarship increasingly challenges the effectiveness and equity of carbon markets as a global climate strategy (Lohmann 2009; Böhm et al. 2012; Bryant 2016). This Policy Analysis contributes by developing a theoretical framework elucidating the paradoxes and fallacies underpinning carbon credit approaches. Drawing on

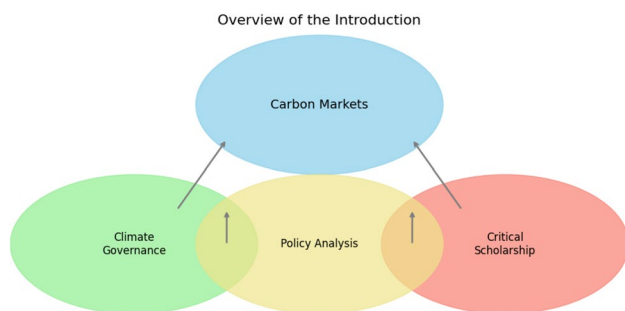


Fig. 1 Overview of the introduction

economics, law, ethics, and political ecology, I argue carbon markets are ineffective at producing real emissions reductions while reproducing environmental injustices, distorting policy priorities, and delaying necessary structural changes.

Section 2 overviews carbon credit systems and their role in global climate governance. Section 3 identifies four key limitations and paradoxes: lack of additionality, exacerbated inequity, cost-effectiveness fallacy, and structural delay. Section 4 discusses implications for transnational climate policy and proposes alternative just transition approaches. Section 5 concludes the article.

2 The Rise of Global Carbon Markets

The idea of using market mechanisms to control pollution emerged in the 1960 s, influenced by economists such as Coase (1960) and Dales (1968). They argued environmental problems result from lacking clearly defined property rights over ecological resources. Establishing tradable property rights could allow market efficiencies to determine optimal pollution levels (Lane 2012). This logic was first applied to carbon dioxide and other GHGs in the early 1990 s (Lohmann 2010).

The primary carbon market instruments are emissions allowances (or permits) and offsets (Bumpus and Liverman 2008). Under cap-and-trade, a central authority sets an emissions cap for regulated sectors and issues a corresponding number of allowances. Companies must surrender enough allowances to cover their emissions or face penalties. Surplus allowances from reducing emissions below the cap can be banked or traded to other companies. Offsets work similarly, but are generated by projects reducing emissions relative to a hypothetical business-as-usual baseline, often in economic sectors or geographic regions not covered by emissions caps (Lovell 2010).

The 1997 Kyoto Protocol marked a key turning point, establishing the first global carbon markets (Viñuales et al. 2021). Kyoto set binding emissions targets for industrialised nations while introducing ‘flexibility mechanisms’ to reduce compliance costs (Ciplet 2015). These included International Emissions Trading between countries with targets and two offset programs: the Clean Development Mechanism (CDM), generating credits from projects in developing countries, and Joint Implementation (JI) for projects in developed nations and economies in transition (Newell and Paterson 2010).

Although the USA failed to ratify Kyoto, carbon markets proliferated in the 2000s. The European Union launched the world’s largest cap-and-trade system in 2005 (EU-ETS), followed by programs in New Zealand, California, Quebec, and South Korea (ICAP 2022). Voluntary markets also expanded as companies and individuals

purchased offsets to demonstrate climate action or ‘carbon neutrality’ (Lovell and Liverman 2010). The number and size of carbon pricing initiatives have increased substantially over the last decade (World Bank 2022).

The Paris Agreement further cemented carbon markets’ role in global climate governance (Kuyper et al. 2018). Article 6 provides for ‘internationally transferred mitigation outcomes’ tradable between countries and a new mechanism to generate offsets for governments and private buyers (Marcu 2016; Cheong 2024). The rules governing these provisions were finalised at the 2021 Glasgow climate conference, enabling post-2020 carbon market expansion (UNFCCC 2021).

Today, carbon is traded as a commodity worth over \$850 billion annually (Ecosystem Marketplace 2022). The carbon credit market size is estimated to increase by over \$1.4 billion at a compound annual growth rate exceeding 30% between 2023 and 2028 (Technavio 2024). However, the growing reach of carbon markets has amplified debates about their environmental integrity and socioeconomic implications (Pearse and Böhm 2014; Bryant 2016). The following sections illustrate how establishing global carbon credit systems has paradoxically undermined climate mitigation efforts in multiple ways.

3 Limitations and Paradoxes of Carbon Markets

3.1 Lack of Additionality

In the context of carbon credit systems, the following concepts illustrate the challenges associated with additionality:

Business as usual emissions	This represents the baseline level of emissions that would occur without any carbon credit projects. It serves as the reference point for evaluating the impact of carbon credit initiatives
Claimed emissions reductions	This line represents the level of emissions reduction claimed by project developers through the implementation of carbon credit projects. These reductions are often portrayed as significant and substantial compared to the baseline

Business as usual emissions	This represents the baseline level of emissions that would occur without any carbon credit projects. It serves as the reference point for evaluating the impact of carbon credit initiatives
Actual emissions reductions	This line indicates the true level of emissions reduction achieved by the project. In many cases, actual reductions are much closer to the business as usual emissions, revealing a smaller impact than claimed
Inflated baselines	These are hypothetical scenarios constructed to justify higher claims of emissions reductions. By inflating the baseline, project developers can present their initiatives as achieving greater reductions than what is actually occurring
Non-additional projects:	These are projects that would have been implemented regardless of the availability of carbon credits. Examples include renewable energy projects that are already economically viable without the need for carbon credit incentives. Such projects do not contribute to additional emissions reductions, undermining the integrity of the carbon credit system

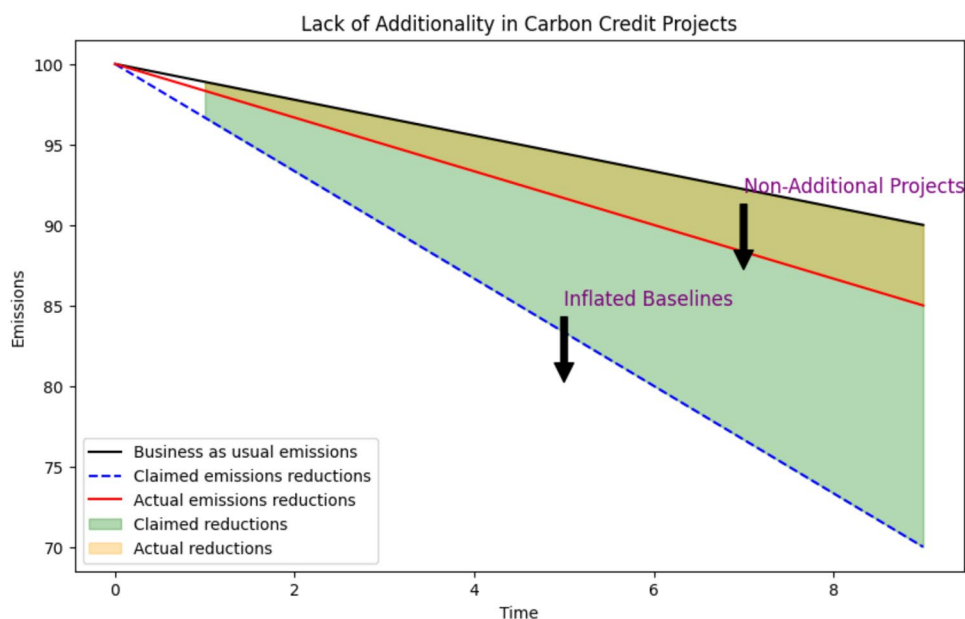
Figure 2 visually illustrates the key challenges associated with the lack of additionality in carbon credit systems. It highlights the discrepancies between claimed and actual emissions reductions, the problem of inflated baselines, and the prevalence of non-additional projects. As depicted,

the ‘Business as usual emissions’ line serves as the baseline scenario without any carbon credit interventions. The ‘Claimed emissions reductions’ line shows the purported reductions by project developers, which often exaggerate the actual impact. The ‘Actual emissions reductions’ line reveals that the true reductions are much closer to the business-as-usual scenario, indicating that many projects fail to deliver the additional benefits they claim. Inflated baselines and non-additional projects further undermine the integrity of these claims. These visual elements underscore the complex and often flawed nature of determining additionality, reinforcing the points made in the following detailed analysis of the issue.

A fundamental assumption of carbon credit systems is they incentivise emissions reductions that would not otherwise occur under business-as-usual conditions (Bushnell 2012). Credits must be ‘additional’ to what would have happened anyway to represent real emissions reductions (Gillenwater 2012). However, there are strong reasons to believe many carbon credits fail this additionality requirement.

The conceptual and practical challenges of determining additionality have been widely discussed for carbon offsets (Bumpus and Liverman 2008; Schneider 2009). Offset credits are calculated based on the difference between project emissions and a hypothetical baseline scenario without the project (Lovell 2010). Constructing these counterfactual baselines inevitably involves high uncertainty and subjectivity (Gillenwater 2012). Studies have documented widespread over-crediting of offsets due to inflated baselines and crediting of non-additional renewable energy and industrial gas projects that would have been implemented regardless (Haya 2009; Cames et al.

Fig. 2 Lack of additionality in carbon credit projects



2016; Calel et al. 2021). In 2023, researchers found most REDD projects have not significantly reduced deforestation (West et al. 2023).

In December 2023, the World Bank announced plans for a high-integrity global carbon market to help 15 resource-rich countries earn income from forest protection carbon credits (World Bank 2023). However, it remains to be seen if this would improve carbon market integrity or avoid durability issues affecting the four largest voluntary market registries (Macfarlane 2022).

Additionality issues also apply to emissions allowances under cap-and-trade. In the Kyoto Protocol and EU-ETS, generous free allocation based on historical emissions (grandfathering) has enabled polluters to amass large allowance surpluses without significant emissions reductions (Rajamani et al. 2021; Ellerman and Buchner 2008; Whitmore 2017). Regulated entities often receive allowances exceeding their actual emissions, generating ‘hot air’ credits not associated with climate benefits (Green 2018). This lack of scarcity undermines the incentive to reduce emissions.

Selling non-additional credits compromises global climate mitigation. When emitters use such credits to ‘offset’ emissions instead of reducing them at the source, the result is a net global emissions increase (Hyams and Fawcett 2013). Unused allowance surpluses accumulated from previous EU-ETS stages when caps were less ambitious. Furthermore, a design flaw permitted excessive offset imports, inflating the surplus excessively (Kuch 2015). As long as oversupply continues, carbon markets will fail to produce real emissions reductions (Assous 2023; Neuhoff 2012).

In May 2023, a class-action lawsuit against a major American airline in California highlighted risks associated with climate-washing. The plaintiffs allege the airline’s assertion of being the first globally carbon-neutral airline is deceptive, primarily depending on questionable low-quality offsets. They contend the airline aimed to benefit financially by deceiving environmentally conscious consumers willing to pay more for eco-friendly services (Greenfield 2023). This case highlights legal risks of junk offsets.

Furthermore, in October 2023, a policy report highlighted corruption and integrity risks in climate solutions as an emerging global challenge (Chan et al. 2023). For example, companies often use carbon credits to offset emissions and claim carbon neutrality, but lack of proper regulation and oversight in carbon markets raises concerns about the effectiveness and legitimacy of these credits. One main problem is the absence of a clear legal framework to prevent false or misleading carbon credit claims. While voluntary corporate action like the VCM Claims Code of Practice (Voluntary Carbon Markets Integrity Initiative 2023) recommends companies only use carbon credits to finance additional mitigation beyond their targets, there is still a lack of transparency and accountability in the carbon credit market. This leaves

room for greenwashing by purchasing low-quality or questionable credits not leading to genuine emission reductions (Chan et al. 2023).

3.2 Exacerbated Inequity

Carbon markets have been promoted as directing finance from wealthy to poorer countries to support climate mitigation and sustainable development (Mehling et al. 2018). In practice, however, global carbon markets and offset projects have often exacerbated international inequities and concentrated environmental risks in marginalised communities.

One key equity concern is carbon markets enable Global North polluters to outsource mitigation responsibilities to the Global South rather than reducing their own emissions (Lohmann 2008; Bachram 2004). By investing in cheap credits from developing countries, companies can continue emitting while purportedly compensating for climate impacts elsewhere. However, this perpetuates uneven power relations and neocolonial resource extraction patterns (Bumpus and Liverman 2011). In the quest for least-cost abatement, offset projects are disproportionately located in regions with limited financial and institutional capacity to regulate environmental integrity (Boyd et al. 2011).

This geographic shift in mitigation produces localised socio-environmental harms. Some of the largest offset credit sources, including industrial gas destruction, hydropower, and biomass energy projects, can negatively impact local communities while failing to generate sustainable development benefits (Böhm and Dabhi 2009; Newell and Bumpus 2012). Indigenous peoples and other frontline communities often face displacement or exclusion for carbon projects, losing access to land and resources (Cabello and Gilbertson 2012; Chomba et al. 2016). The extensive history of human rights abuses linked to offset projects was a key reason for social safeguards in Article 6 of the Paris Agreement (Webb and Wentz 2018; Obergassel et al. 2017).

Carbon markets can also entrench inequality through new forms of exclusion and impoverishment. Governments have used establishing forest carbon offsets to assert control over rural land, criminalising traditional practices such as shifting cultivation and increasing food insecurity (Asiyanbi 2016; Osborne and Shapiro-Garza 2018). Poor households are excluded from benefiting from carbon finance when offset methodologies favour large-scale plantations and strict forest enclosures over smallholder systems (Cavanagh and Benjaminsen 2014; Chomba et al. 2016). By perpetuating ‘green grabbing’ (Fairhead et al. 2012), carbon markets can reproduce environmental injustices.

Despite generating immense profits for fossil fuel corporations and financial intermediaries, carbon market revenues have largely failed to reach poor Global South communities (Lohmann 2008). The CDM has been critiqued for its poor

sustainable development record, with revenues unlikely to benefit host communities (Hultman et al. 2020; Sutter and Parreño 2007). Although carbon markets have created vast opportunities for capital accumulation, benefits accrue primarily to polluters and market makers rather than frontline climate change communities (Barros and Wilk 2021).

3.3 Cost-Effectiveness Fallacy

The bar chart in Fig. 3 illustrates the cost-effectiveness fallacy in carbon markets. The current focus on low-cost mitigation options, represented by the red bars, aligns with the argument that carbon markets, by design, channel finance towards the cheapest abatement opportunities (Broekhoff et al. 2019). However, this short-term cost minimisation approach comes at the expense of long-term emission reductions. The blue bars, indicating the needed focus, emphasise the necessity to redirect investment towards essential long-term options and deep decarbonisation, even if they are initially costlier (Lohmann 2009). The chart visually reinforces the assertion that the myopic focus on cost containment distorts broader climate policy priorities and weakens mitigation ambition (Pearse and Böhm 2014). It underscores the need to look beyond the least-cost abatement and prioritise options that enable the structural changes and technological innovation required for a stable climate future (Böhm et al. 2012).

A core carbon market rationale is they minimise the social cost of climate mitigation by directing abatement to where it is cheapest (Stern 2007; Gollier and Tirole 2015). This cost-effectiveness framing has become the dominant logic shaping climate policy choices (Parson and Kravitz 2013). However, the preoccupation with short-term cost minimisation comes at the expense of long-term emission reductions, representing a false economy.

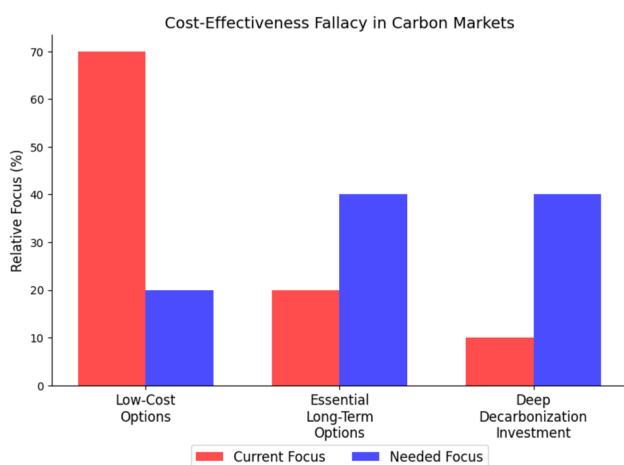


Fig. 3 Cost-effectiveness fallacy in carbon markets

By design, carbon markets channel finance toward the lowest-cost abatement options such as industrial efficiency, waste methane capture, and preventing deforestation (Broekhoff et al. 2019). While such activities can yield meaningful emissions reductions, they have inherent physical and economic limitations. Only a finite number of such ‘low-hanging fruit’ exist before opportunities for further reductions become exhausted or prohibitively expensive (Grubb et al. 2014). Relying primarily on offsetting therefore delays economy-wide decarbonisation and fails to incentivise the technological innovation needed for deep emissions cuts (Wara 2007; Wara and Victor 2008).

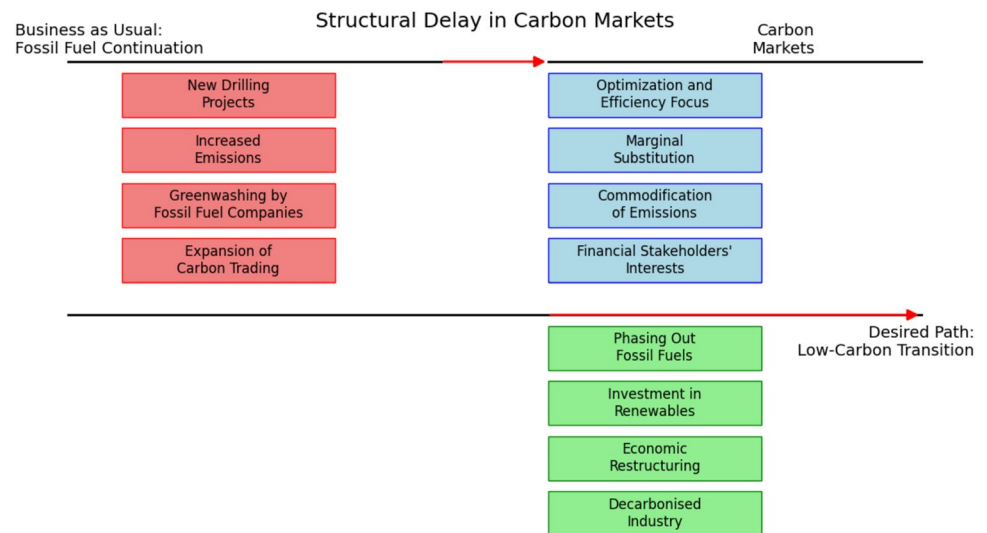
Equally problematic is carbon markets direct investment away from costlier but essential long-term solutions such as renewable energy deployment and transportation electrification (Lohmann 2009). Offset market revenues are rarely sufficient to support transformative low-carbon infrastructure, as profit-seeking project developers will always favour the cheapest possible interventions (Wara 2007). This dynamic is evident in the CDM experience, which generated huge volumes of cheap industrial gas offsets but failed to mobilise significant renewable energy investment in developing countries (Schneider 2009; Victor 2009).

The myopic focus on short-term cost containment distorts broader climate policy priorities and weakens mitigation ambition (Pearse and Böhm 2014). Carbon prices consistently remain far below levels needed to incentivise deep decarbonisation (Edenhofer et al. 2017; World Bank 2022). Policymakers have shied away from imposing stringent emissions caps or reforming market oversupply for fear of raising compliance costs for regulated entities (Green 2021; Narassimhan et al. 2018). In the name of cost minimisation, critical programmatic decisions are shaped by powerful industrial interests rather than scientific imperatives (Lohmann 2012).

Over-reliance on carbon markets as a regulation substitute impedes the structural changes needed to address climate change (Böhm et al. 2012). By emphasising abatement costs rather than ecological stability, market mechanisms obscure the urgency of near-term emissions reductions indicated by climate science (Anderson and Bows 2011). Carbon markets have become not just a means of minimising mitigation costs but of rationalising delays in climate action that jeopardise a stable climate future (Lohmann 2009).

3.4 Structural Delay

Figure 4 illustrates the structural delay in carbon markets. The ‘Business as Usual’ path, representing the continuation of fossil fuels, is diverted towards carbon markets, which are influenced by delaying factors such as optimisation and efficiency focus, marginal substitution, commodification of emissions, and financial stakeholders’ interests (Broekhoff

Fig. 4 Structural delay in carbon markets

et al. 2019; Lohmann 2009). These factors, shown in the blue boxes, contribute to the consequences of the fossil fuel continuation, depicted in the red boxes, including new drilling projects, increased emissions, greenwashing by fossil fuel companies, and the expansion of carbon trading (Böhm et al. 2012). The desired path, the low-carbon transition, is represented at the bottom of the diagram, with actions such as phasing out fossil fuels, investment in renewables, economic restructuring, and decarbonised industry shown in the green boxes. The diagram visually reinforces the argument that the cost-effectiveness fallacy in carbon markets distorts climate policy priorities and delays the necessary structural changes for a stable climate future (Pearse and Böhm 2014).

The most fundamental carbon market paradox is they impede the very economic transformations they are ostensibly intended to facilitate. By allowing continued fossil fuel exploitation in the guise of climate action, carbon credit systems delay the structural shifts needed for a low-carbon transition (Lohmann 2012; Pearse and Böhm 2014).

Carbon markets are premised on economic principles of optimisation, efficiency, and marginal substitution (Lohmann 2005). They rely on assumptions that carbon pollution is a quantifiable externality that can be internalised through pricing and that fossil and non-fossil energy sources are largely fungible and interchangeable (Newell and Paterson 2010; Lane 2012). In this framing, climate change becomes little more than a market failure to be corrected through the right price signals (Bryant 2016).

However, this market logic is incommensurable with the scale of economic restructuring needed for deep decarbonisation (Mann and Wainwright 2018). Avoiding dangerous climate change will require not just incremental emissions reductions but a wholesale phaseout of fossil fuel production and consumption in the coming decades (McGlade and Ekins 2015). It is not possible to simply 'optimise' our way

out of carbon lock-in when fossil fuel extraction and combustion are so deeply embedded in nearly all facets of industrial economies (Unruh 2000).

Carbon markets have proven ineffective at catalysing this structural shift away from fossil fuels. The revenues they generate flow primarily to projects that tinker at the margins of industrial production rather than disrupting the dominance of fossil capital (Lohmann 2008). Despite decades of carbon market activity, fossil fuel production and combustion continue expanding globally, locking in future emissions growth (Hickel and Kallis 2020). Emissions dipped in 2020 due to pandemic-related slowdowns but quickly rebounded to record highs in 2021 (IEA 2021).

Perversely, carbon markets have provided legitimacy and new profit opportunities for industries most responsible for perpetuating fossil fuel extraction (Newell and Paterson 2010). Oil and gas companies have been among the largest buyers and traders of carbon credits, using offsetting to justify expanded drilling and greenwash their public image (Lohmann 2009; Sapinski 2015). Fossil fuel corporations have lobbied extensively for using international offsets in climate policy to protect their core business models (Meckling 2011; Ferns et al. 2019).

Carbon markets have also created new vested interests in the continued commodification of the atmosphere (Pearse and Böhm 2014). A vast array of banks, brokers, exchanges, verifiers, and project developers now have a direct financial stake in the ongoing expansion of carbon trading (Lohmann 2008; Knox-Hayes 2016; Somosi et al. 2024). The emergence of this 'climate services' sector has generated a powerful constituency benefiting from intensifying offsetting, regardless of its environmental outcomes (Newell and Paterson 2010).

By entrenching fossil capital and financialising the climate crisis, carbon markets inhibit the structural changes

and political-economic realignments needed for an equitable post-carbon transition (Newell and Simms 2020). They reproduce the very power relations that have generated the climate crisis, delaying systemic solutions (Lohmann 2012). As long as carbon credit systems offer an outlet for continued fossil fuel production and a new arena for financial accumulation, they will serve as an impediment to real climate progress.

4 Implications for Climate Policy and Governance

The preceding analysis highlights the need for a fundamental rethinking of global carbon credit frameworks. As currently structured, carbon markets not only fail to produce emissions reductions but actively undermine efforts to address climate change. Transcending the limitations of carbon markets will require a multi-pronged approach centred on equity, ecological effectiveness, and deep decarbonisation. Key elements could include:

- (a) Accelerating fossil fuel phaseouts: Carbon markets must transition away from offsetting and toward the managed decline of fossil fuel production (Green 2018). This could involve policies such as fossil fuel subsidy reform, bans on new extraction and infrastructure, and just transition programs for affected workers and communities (Trout et al. 2022).
- (b) Scaling up public investment: Rather than relying on unstable carbon markets, the climate finance needed for a low-carbon transition should be mobilised through public sources such as progressive taxation, redirection of military spending, and monetary financing (Monbiot 2023; Demertzis et al. 2024). Public investment can catalyse transformative infrastructure projects and overcome the short-term cost constraints of private finance.
- (c) Advancing climate justice: Equity and human rights must be central to emissions reduction efforts, not treated as an afterthought (Schlosberg and Collins 2014). Principles of climate justice include prioritising indigenous sovereignty, respecting community consent for mitigation projects, and channelling resources to frontline communities experiencing climate harms (Seck 2017; van Asselt and Zelli 2018).
- (d) Rethinking carbon accounting: Emissions targets and accounting frameworks should shift away from market-based offsetting and toward aggregate emissions reductions from a zero baseline (Gunderson et al. 2020). This could involve consumption-based accounting to address international emissions transfers (Davis and Caldeira 2010), as well as better tracking of emissions from land use change (Houghton et al. 2012).
- (e) Implementing cap-and-dividend: An alternative approach is to replace carbon markets entirely with a cap-and-dividend system (Barnes 2014; Kunkel and Kammen 2011). Under this model, a stringent cap would be placed on fossil fuel extraction and imports at the point of production. Fossil fuel companies would be required to purchase permits at auction, with the revenues redistributed directly to citizens as a universal dividend. This would effectively price carbon while ensuring an equitable distribution of revenues (Boyce 2018). As the cap declines over time, clean energy would become increasingly cost-competitive, driving a rapid transition away from fossil fuels (Fremstad and Paul 2019). This transition would be facilitated by public investment in low-carbon infrastructure, funded by diverting subsidies from fossil fuels to clean energy (Jacobson et al. 2022). Cap-and-dividend has several advantages over carbon markets: it ensures emissions reductions are achieved domestically, generates public buy-in through dividend payments, and avoids the speculation and volatility associated with carbon trading (Cullenward and Victor 2020). Implementing cap-and-dividend would require overcoming political barriers, including opposition from fossil fuel interests (Green 2021). However, it offers a pathway to more effective and equitable climate mitigation than carbon markets can provide.

Operationalising these principles will require new multi-scalar and polycentric governance arrangements (Jordan et al. 2018). This could include, for example, linking emissions trading systems with declining caps to border carbon adjustments and international technology transfer mechanisms (Mehling et al. 2018). Voluntary cooperation ‘climate clubs’ among high-ambition states could also help accelerate the transition (Falkner et al. 2022).

Ultimately, however, a stable climate future will require more transformative economic and political paradigms beyond the market-based status quo (Hickel 2020). Alternative frameworks such as degrowth (Kallis et al. 2018), eco-socialism (Löwy 2015), and energy democracy (Burke and Stephens 2018) offer visions for decarbonisation that centre sufficiency, justice, and collective governance. Overreliance on carbon markets risks foreclosing these possibilities and entrenching a system incapable of achieving climate stability (Mathai et al. 2018).

5 Conclusion

This Policy Analysis has explored the problem through a theoretical framework elucidating the paradoxes and limitations of carbon markets as a global climate mitigation strategy. I have argued that carbon credit systems not only fail to produce real emissions reductions, but also serve to exacerbate inequities, distort policy priorities, and delay the structural changes needed for deep decarbonisation.

The rapid expansion of carbon markets in recent decades represents a monumental sociopolitical experiment with little empirical basis or democratic oversight (MacKenzie 2009). Given their manifest environmental ineffectiveness and deleterious social consequences, the continued use of international carbon markets as a centrepiece of climate policy is ethically and ecologically unjustifiable. The urgency of the climate crisis demands proven solutions that can catalyse real-world emissions reductions while advancing equity and justice.

Moving beyond the false promises of carbon markets will require a far-reaching transformation of the structures of capital accumulation, production, and consumption that define the global political economy (Pattberg and Zelli 2017). While complex, this low-carbon transition is both necessary and achievable. By phasing out fossil fuels, mobilising public investment, directing frontline communities, and rethinking carbon governance, it is possible to build a more stable and just climate future.

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Declarations

Conflict of interest The author has no competing interests to declare that are relevant to the content of this article.

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